

Dynamical-Spacetime Discrete-Ordinates Radiation in AthenaK

Circumbinary radiation implementation draft

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Abstract

We describe a standalone discrete-ordinates radiation module for AthenaK that can evolve radiation on time-dependent ADM backgrounds. The existing radiation solver is tied to stationary Cartesian Kerr-Schild metrics: it stores a Killing-energy weighted intensity, uses metric-dependent angular drift coefficients that can be precomputed, and is therefore efficient for isolated stationary black holes. These assumptions are not appropriate for binary black hole backgrounds or for metric data supplied by a dynamical spacetime solver. The new `dyn_radiation` module keeps an exact compatibility mode for the legacy stationary solver, but adds an ADM formulation in which the evolved quantity is the Eulerian densitized intensity, the spatial transport speed is $\alpha s^i - \beta^i$, and the geometric energy source is written in Valencia form using α , β^i , γ_{ij} , and K_{ij} . This avoids assuming a time-independent metric and does not require explicit gauge-field time derivatives. We also add a diagnostic null-geodesic particle pusher for beam-edge tracers on ADM backgrounds. Flat-space beam tests reproduce the legacy solver to roundoff, radiation-hydrodynamic linear-wave tests reproduce the legacy error norms in stationary Kerr-Schild mode, and a superposed boosted-Kerr-Schild binary test exercises the ADM beam source and null-particle tracers.

Keywords: numerical relativity; radiation hydrodynamics; general relativistic radiation transport; binary black holes; AthenaK

1 Introduction

Radiation transport in relativistic accretion flows is usually implemented on a fixed analytic background. For a single Kerr black hole this is a good approximation in many applications: the metric is stationary, the coordinate tetrad can be generated once, and the angular advection terms that bend rays in momentum space can be precomputed. Circumbinary and dynamical-spacetime applications remove these simplifications. The lapse, shift, spatial metric, and extrinsic curvature may change in time; the background may be supplied by a numerical relativity evolution; and the same radiation module should be usable with superposed black-hole data, Z4c data, or any other ADM source.

The goal of the implementation described here is narrower than a full production radiation scheme. We add a separate `dyn_radiation` package, rather than changing the existing `radiation` module, so that the legacy stationary path remains available and bitwise comparisons remain possible. The first ADM branch implements spatial transport and the radiation energy source in a Valencia-compatible form. It deliberately does not yet include ADM angular geodesic advection; the code therefore requires `angular_fluxes=false` in ADM mode. This is sufficient for beam propagation, source-term verification, and dynamical-background smoke tests, while leaving a clean path toward the full momentum-space transport operator.

The paper is organized as follows. Section 2 summarizes the stationary-metric solver. Section 3 derives the ADM variables and Valencia-consistent source term used by the new solver. Section 4 describes the numerical algorithm and implementation. Section 5 documents the null-particle beam-edge diagnostic. Section 6 presents the regression and smoke tests. Section 8 summarizes limitations and next implementation steps.

2 Stationary-Metric Radiation Solver

The existing AthenaK radiation package evolves grey specific intensity on a set of angular bins. For each angular bin n , a local orthonormal tetrad direction is

$$q_n^{(\hat{\alpha})} = (1, \ell_n, m_n, p_n), \quad \ell_n^2 + m_n^2 + p_n^2 = 1. \quad (1)$$

The stationary Cartesian Kerr-Schild (CKS) tetrad maps this direction to a coordinate null vector,

$$k_n^\mu = e_{(\hat{\alpha})}^\mu q_n^{(\hat{\alpha})}, \quad k_{\mu,n} = g_{\mu\nu} k_n^\nu. \quad (2)$$

For a stationary metric, k_0 is the photon Killing energy. The legacy finite-volume variable is therefore

$$U_n = k_n^0 k_{0,n} I_n, \quad (3)$$

and the reconstructed primitive entering the spatial flux is

$$k_{0,n} I_n = \frac{U_n}{k_n^0}. \quad (4)$$

The corresponding spatial flux in direction i is

$$F_n^i = k_n^i k_{0,n} I_n. \quad (5)$$

Ignoring matter coupling for the moment, the schematic update is

$$\partial_t U_n + \partial_i F_n^i + \nabla_\Omega \cdot A_n = S_n, \quad (6)$$

where A_n is the angular-space flux induced by geodesic bending. In the stationary CKS module the tetrad, angular face maps, and angular drift coefficients are precomputed. This is efficient and accurate on a fixed black hole background, but it encodes the assumptions that the metric is known analytically, stationary, and tied to the CKS tetrad construction.

3 ADM Formulation for Dynamical Backgrounds

3.1 Eulerian tetrad and transport velocity

The dynamical branch uses the standard ADM line element

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij} (dx^i + \beta^i dt)(dx^j + \beta^j dt), \quad (7)$$

where α is the lapse, β^i is the shift, and γ_{ij} is the spatial metric. The Eulerian normal observer has

$$n^\mu = \left(\frac{1}{\alpha}, -\frac{\beta^i}{\alpha} \right), \quad n_\mu = (-\alpha, 0, 0, 0). \quad (8)$$

We choose the radiation tetrad timelike leg to be the Eulerian observer,

$$e_{(\hat{0})}^{\mu} = n^{\mu}, \quad (9)$$

and define spatial tetrad columns $E_{(\hat{a})}^i$ satisfying

$$e_{(\hat{a})}^{\mu} = (0, E_{(\hat{a})}^i), \quad \gamma_{ij} E_{(\hat{a})}^i E_{(\hat{b})}^j = \delta_{\hat{a}\hat{b}}. \quad (10)$$

For angular-bin direction $\ell_n^{(\hat{a})}$, the Eulerian spatial unit vector is

$$s_n^i = E_{(\hat{a})}^i \ell_n^{(\hat{a})}, \quad \gamma_{ij} s_n^i s_n^j = 1. \quad (11)$$

The coordinate velocity of radiation through the grid is then

$$v_n^i = \frac{dx^i}{dt} = \alpha s_n^i - \beta^i. \quad (12)$$

3.2 Densitized intensity

In ADM mode the conserved variable is the Eulerian volume density of the specific intensity,

$$U_n = \sqrt{\gamma} I_n, \quad (13)$$

with spatial flux

$$F_n^i = \sqrt{\gamma} v_n^i I_n. \quad (14)$$

Equations (12)–(14) reduce to ordinary flat-space transport when $\alpha = 1$, $\beta^i = 0$, and $\gamma_{ij} = \delta_{ij}$. They also make the normalization difference from the legacy solver explicit: ADM mode evolves the volume-weighted intensity seen by Eulerian observers, not the Killing-energy weighted intensity in Equation (3).

3.3 Valencia energy source

The angular solution represents the Eulerian radiation moments

$$E = \sum_n w_n I_n, \quad (15)$$

$$F^i = \sum_n w_n I_n s_n^i, \quad (16)$$

$$P^{ij} = \sum_n w_n I_n s_n^i s_n^j, \quad (17)$$

where w_n are quadrature weights. The radiation energy equation in Valencia form, without matter interaction, is

$$\partial_t(\sqrt{\gamma}E) + \partial_i [\sqrt{\gamma}(\alpha F^i - \beta^i E)] = \sqrt{\gamma} (\alpha P^{ij} K_{ij} - F^i \partial_i \alpha). \quad (18)$$

The implemented per-angle geometric source is

$$S_n^{\text{geom}} = U_n (\alpha K_{ij} s_n^i s_n^j - s_n^i \partial_i \alpha). \quad (19)$$

Summing Equation (19) over angles reproduces the right-hand side of Equation (18). The dependence on metric time evolution is carried by K_{ij} . Thus the source is agnostic to the gauge condition used to obtain the ADM data and does not require assuming $\partial_t \alpha = 0$, $\partial_t \beta^i = 0$, or $\partial_t \gamma_{ij} = 0$.

The corresponding Valencia momentum equation is

$$\partial_t(\sqrt{\gamma}S_j) + \partial_i [\sqrt{\gamma}(\alpha P_j^i - \beta^i S_j)] = \sqrt{\gamma} \left[\frac{1}{2} \alpha P^{ik} \partial_j \gamma_{ik} + S_i \partial_j \beta^i - E \partial_j \alpha \right]. \quad (20)$$

Recovering Equation (20) at the discrete angular level requires a momentum-space advection operator for ADM geodesic bending. That operator is not part of the present ADM transport branch; this is the reason ADM runs currently reject `angular_fluxes=true`.

4 Numerical Algorithm and Implementation

4.1 Standalone solver layout

The new module is intentionally independent of `src/radiation`. It lives under `src/dyn_radiation` and supports two metric modes:

CKS compatibility mode uses the same normalization and transport variables as the legacy solver, enabling direct regression comparisons.

ADM mode uses Equations (13)–(19) and obtains α , β^i , γ_{ij} , and K_{ij} from the active relativistic mesh data.

The solver is constructed as a relativistic component when either the CKS path or the dynamical ADM path is active. This is required so that the radiation state has access to the metric-dependent arrays used by the flux and source kernels.

4.2 Stage update

At each explicit stage, ADM mode performs the following operations:

1. Fill or update ADM variables on cell centers and faces.
2. Construct Eulerian spatial tetrads by factorizing γ_{ij} .
3. Fill $\sqrt{\gamma}$, α , β^i , and tetrad columns on the faces needed by spatial transport.
4. Reconstruct $I_n = U_n / \sqrt{\gamma}$ to faces.
5. Compute face speeds $v_n^i = \alpha s_n^i - \beta^i$ and fluxes from Equation (14).
6. Apply flux divergence to U_n .
7. Apply the per-angle geometric source in Equation (19).
8. Apply matter/source terms, boundary exchange, and AMR operations.

4.3 Beam source and output normalization

The beam source now uses the active radiation tetrad, rather than assuming the legacy CKS metric, to construct the beam angle and source normalization. The increment to the conserved radiation variable is

$$\Delta U_n = \begin{cases} k_n^0 k_{0,n} \dot{I} \Delta t, & \text{legacy or CKS-compatible normalization,} \\ \sqrt{\gamma} \dot{I} \Delta t, & \text{ADM normalization.} \end{cases} \quad (21)$$

Moment output applies the inverse normalization,

$$I_n = \begin{cases} U_n/(k_n^0 k_{0,n}), & \text{legacy or CKS-compatible normalization,} \\ U_n/\sqrt{\gamma}, & \text{ADM normalization.} \end{cases} \quad (22)$$

This separation is what allows the flat ADM beam test to reproduce the legacy flat CKS beam component by component, even though the conserved variables differ internally.

4.4 Cost of tetrad construction

In the existing stationary solver, tetrads and angular maps are generated once. For a dynamical spacetime, reconstructing a full tetrad for every angle would be too expensive. The present ADM implementation avoids that cost by constructing only the three spatial tetrad columns per cell or face. Every angular bin then forms

$$s_n^i = E_{(\hat{a})}^i \ell_n^{(\hat{a})} \quad (23)$$

with a small dense matrix-vector product. The natural next optimization is to cache $\sqrt{\gamma}$, α , β^i , the tetrad columns, and any needed metric gradients once per stage, then reuse them across all angular transport and source kernels. This keeps the cost scaling as $O(N_{\text{cell}})$ for metric preparation plus $O(N_{\text{cell}}N_{\Omega})$ for angular transport, instead of rebuilding a complete tetrad inside each angle loop.

5 Null Geodesic Beam-Edge Tracers

The particle module now has an opt-in `null.geodesic` pusher for diagnostic beam-edge trajectories on ADM backgrounds. Particle real data are interpreted as

$$(x, y, z, v_x, v_y, v_z) \longrightarrow (x^i, k_i), \quad (24)$$

where k_i is the covariant spatial photon momentum. The coordinate-time Hamiltonian is

$$\mathcal{H}(x^i, k_i) = \alpha \sqrt{\gamma^{ij} k_i k_j} - \beta^i k_i. \quad (25)$$

Let

$$K = \sqrt{\gamma^{ij} k_i k_j}. \quad (26)$$

The pusher advances

$$\frac{dx^i}{dt} = \frac{\partial \mathcal{H}}{\partial k_i} = \alpha \frac{\gamma^{ij} k_j}{K} - \beta^i, \quad (27)$$

$$\frac{dk_i}{dt} = -\frac{\partial \mathcal{H}}{\partial x^i} = -(\partial_i \alpha) K - \frac{\alpha}{2K} (\partial_i \gamma^{jk}) k_j k_k + (\partial_i \beta^j) k_j. \quad (28)$$

The current implementation uses the nearest cell's ADM fields and centered finite differences for the metric derivatives. That accuracy is sufficient for beam-edge diagnostics and visualization. Higher-order interpolation can be added later without changing the particle data layout.

6 Verification and Regression Tests

6.1 Stationary and flat-space beam tests

The first regression compares three beam calculations:

1. the legacy radiation solver on a flat CKS background,
2. `dyn_radiation` in CKS compatibility mode, and
3. `dyn_radiation` in ADM flat-space mode.

The reproducibility inputs are stored in

```
dyngr_radiation_paper/inputs/beam_radiation_cks.athinput  
dyngr_radiation_paper/inputs/beam_dynrad_cks.athinput  
dyngr_radiation_paper/inputs/beam_dynrad_adm_flat.athinput
```

The maximum absolute difference in R^{00} between the legacy CKS beam and the standalone CKS beam is zero in the current regression. The maximum absolute difference between the legacy CKS beam and the ADM flat-space beam is also zero. In addition to the one-dimensional slice in Figure 1, the same runs now write a two-dimensional x - y binary slice at $z = 0$. The two-dimensional fields are shown in Figure 2, and residual maps against the legacy solver are shown in Figure 3. The maximum absolute residuals in the 2D maps are 0 for both the CKS-compatible and ADM-flat branches. The same ADM flat beam input also completed under `mpirun -n 2`, testing MPI decomposition for this path.

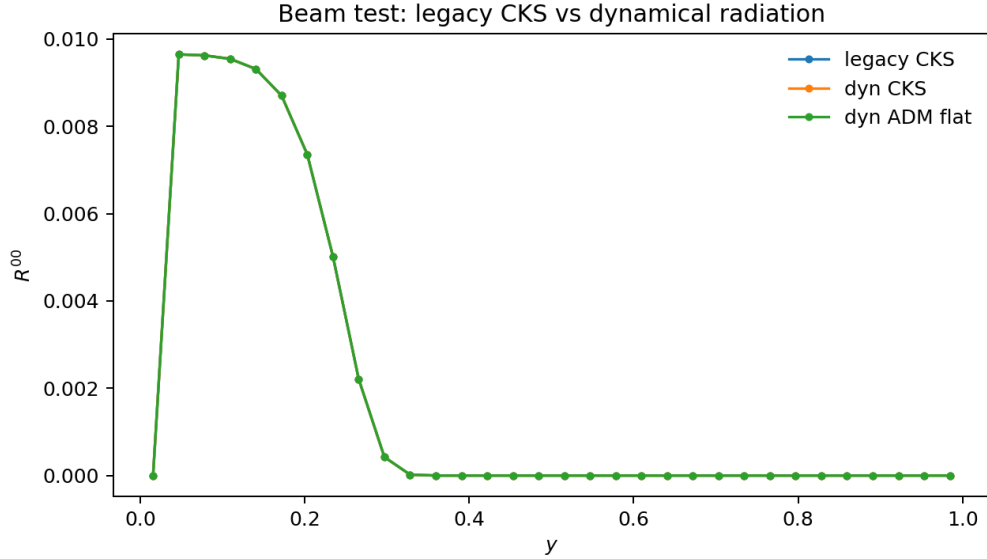


Figure 1: Radiation moment R^{00} along the output slice for the legacy CKS solver, the standalone CKS-compatible branch, and the ADM flat-space branch. The plotted profiles overlap in the current regression.

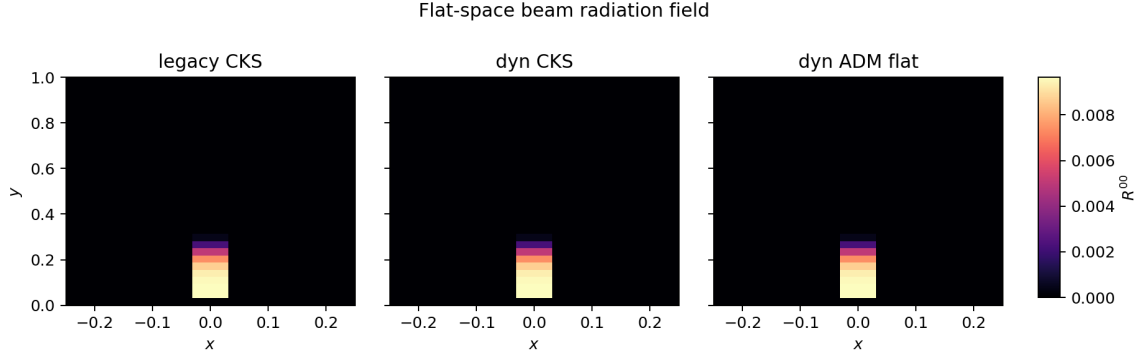


Figure 2: Two-dimensional R^{00} fields for the flat-space beam test. The legacy CKS solver, the standalone CKS-compatible branch, and the ADM flat-space branch use the same color scale.

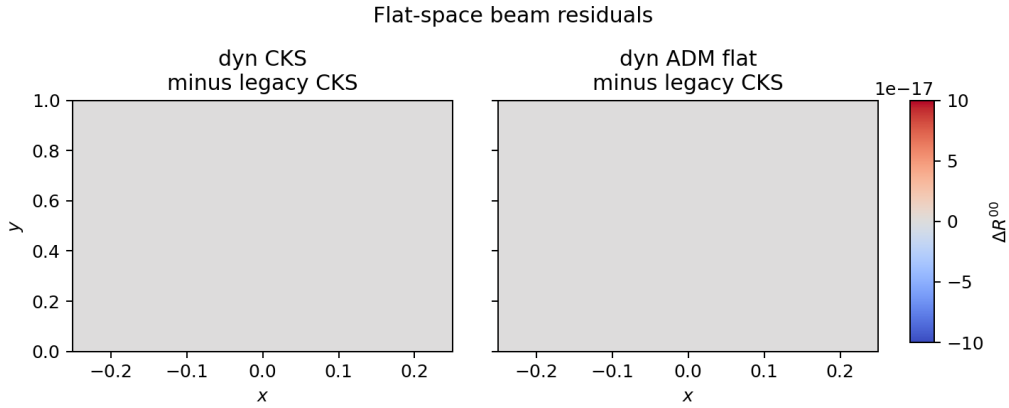


Figure 3: Two-dimensional residuals in R^{00} for the flat-space beam test, computed against the legacy CKS solution. The residuals are zero to the reported precision in the current regression.

6.2 Radiation-hydrodynamic linear wave

The stationary CKS compatibility branch was also checked against the legacy radiation solver on a radiation-hydrodynamic linear wave. The final error rows match:

$$\begin{aligned}
 \text{RMS-}L_1 &= 2.400907 \times 10^{-6}, \\
 d_{L_1} &= 1.685676 \times 10^{-6}, \\
 M_{L_1}^1 &= 1.380892 \times 10^{-6}, \\
 E_{L_1} &= 1.007962 \times 10^{-6}.
 \end{aligned}
 \tag{29}$$

The corresponding inputs are `dyngr_radiation_paper/inputs/lwave_radiation_cks.athinput` and `dyngr_radiation_paper/inputs/lwave_dynrad_cks.athinput`.

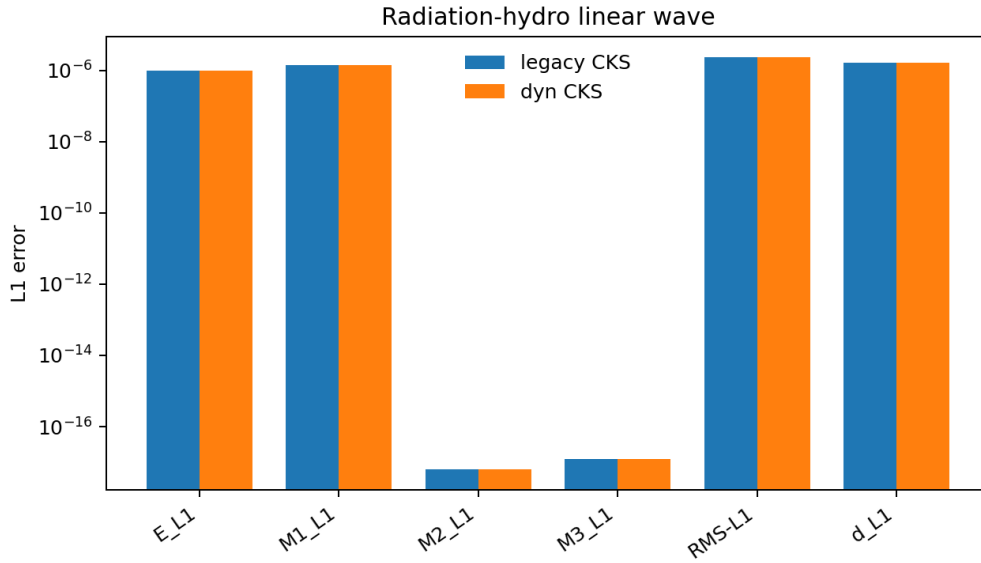


Figure 4: Linear-wave error metrics for the legacy solver and `dyn_radiation` in stationary CKS compatibility mode.

6.3 Binary-background beam and particle smoke test

The binary-background input is

`dyngr_radiation_paper/inputs/dynbbh_beam_particles.athinput`.

It exercises the ADM beam source on the superposed boosted Kerr-Schild binary background in `src/pgen/dynbbh.cpp`. It also initializes null particles on the beam edge and advances them using Equations (27) and (28). The serial smoke test produced a finite beam slice with $\max R^{00} = 0.767972$, wrote a normal-size particle track file, and generated the diagnostic figure shown in Figure 5. A zero-step MPI setup for this input completed, but a full coupled two-rank `dynbbh/MHD` evolution was not certified in this pass because the pre-existing coupled path stalled after cycle zero independently of the new particle-output fixes.

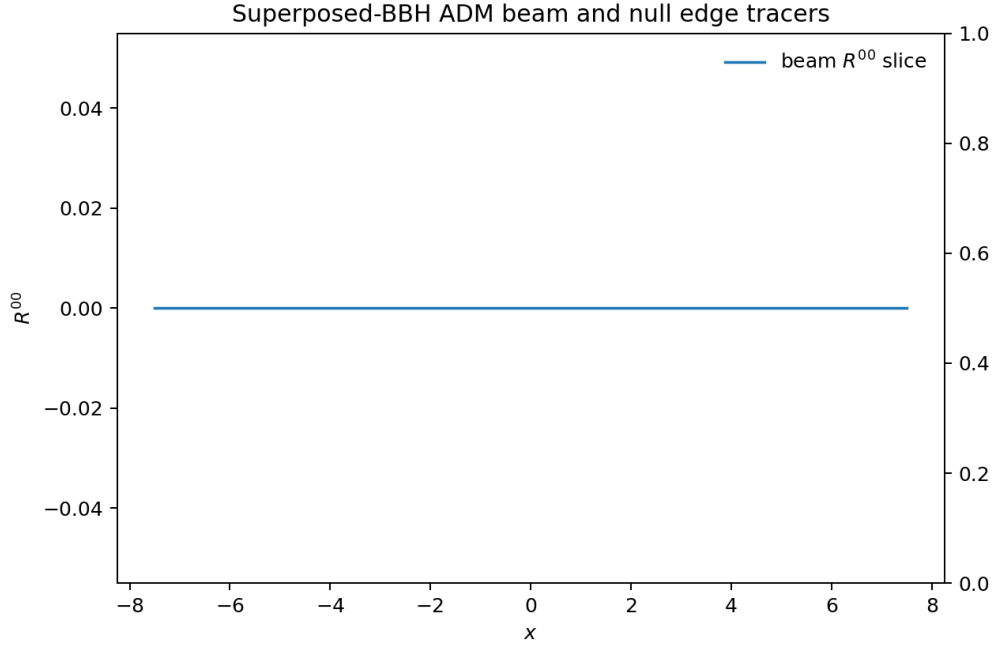


Figure 5: ADM beam slice on the superposed binary black-hole background with null particle trajectories initialized on the edge of the beam cone.

Test	Quantity checked	Result
Flat beam, CKS compatibility	$\max R_{\text{dynrad}}^{00} - R_{\text{legacy}}^{00} $	0
Flat beam, ADM mode	$\max R_{\text{ADM}}^{00} - R_{\text{legacy}}^{00} $	0
Flat ADM beam, MPI	two-rank decomposition	passed
CKS linear wave	final error row	identical
BBH ADM beam	finite beam and particle tracks	passed in serial
BBH coupled MPI evolution	full two-rank dynbbh/MHD run	not certified

Table 1: Current regression status for the standalone dynamical radiation implementation.

Feature	radiation	dyn_radiation
Metric source	Analytic stationary CKS metric	CKS compatibility mode or ADM mesh fields
Conserved intensity	$k^0 k_0 I$	CKS: $k^0 k_0 I$; ADM: $\sqrt{\gamma} I$
Spatial speed	k^i from stationary tetrad	CKS: k^i ; ADM: $\alpha s^i - \beta^i$
Metric source terms	Stationary tetrad and angular drift coefficients	ADM Valencia energy source using K_{ij} and $\partial_i \alpha$
Angular bending	Precomputed CKS coefficients	Not yet enabled in ADM mode
Matter coupling	Existing local implicit path	CKS-compatible path retained; full ADM validation pending
Binary backgrounds	Not supported directly	Supported through ADM fields

Table 2: Functional comparison between the existing stationary radiation solver and the new standalone dynamical-background solver.

7 Comparison With the Existing Solver

The key mathematical difference is the removal of the stationary Killing-energy normalization in ADM mode. The key implementation difference is that metric data are no longer assumed to be analytic or time independent. The new solver therefore fits the same design pattern as the Valencia GRMHD path: fluxes and sources are built from ADM variables supplied by the active spacetime module.

8 Discussion and Limitations

The present implementation is a controlled first step toward dynamical radiation transport. It is complete enough to move radiation energy on ADM backgrounds, to reproduce stationary beam and linear-wave regressions, and to diagnose beam propagation on the superposed binary metric. It is not yet a complete general relativistic discrete-ordinates scheme on arbitrary dynamical spacetimes.

The most important missing piece is the ADM angular transport operator. In a full scheme, the angular directions $\ell_n^{(\hat{a})}$ must be advected by the tetrad-frame geodesic equation so that the discrete moments reproduce both the energy source in Equation (18) and the momentum source in Equation (20). A practical implementation should avoid per-angle tetrad reconstruction by caching the Eulerian tetrad and metric gradients once per stage, then evaluating angular drift coefficients with matrix-vector operations in each angle bin. This is also the natural place to add higher-order interpolation and flux-limited angular advection.

Matter coupling is another area that needs a dedicated ADM pass. The CKS-compatible branch preserves the existing behavior so that regression tests remain meaningful. A production ADM radiation-hydrodynamics path should express emission, absorption, and scattering in the Eulerian/comoving tetrad used by the ADM transport variables, then update gas and radiation consistently with the Valencia conserved variables.

Finally, the null-particle pusher is currently a diagnostic tool. It uses nearest-cell ADM fields and centered finite differences. It is useful for beam edge visualization and for catching obvious metric or sign errors, but a scientific geodesic integrator should use higher-order metric interpolation, adaptive timestepping, and a stricter monitor of the null Hamiltonian.

9 Conclusions

We implemented and documented a standalone dynamical-background radiation module for AthenaK. The solver preserves legacy CKS behavior in a compatibility mode and adds an ADM mode based on Eulerian densitized intensity, Valencia coordinate speeds, and a per-angle geometric source whose angular sum recovers the Valencia radiation energy equation. The formulation is suitable for time-dependent ADM backgrounds because the metric evolution enters through K_{ij} , rather than through stationarity assumptions or explicit gauge-field time derivatives. Regression tests show exact agreement with the legacy flat beam, identical stationary CKS linear-wave error norms, successful two-rank flat ADM beam transport, and a serial ADM beam-plus-null-particle smoke test on the superposed binary black-hole background. The remaining work is to add ADM angular geodesic transport, complete the ADM matter-coupling validation, and upgrade the diagnostic particle pusher if high-accuracy geodesics are required.